



Rajshahi University of Engineering and Technology

Graph-SE-ResUnet: An Approach to Aggregate Multimodal Information

“ Seminar on Thesis Presentation

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Motivation

Motivation

Clinical Need for Automated Tumor Segmentation

- **Gliomas** are among the most aggressive brain tumors, requiring accurate volumetric segmentation for surgical planning and radiotherapy response monitoring.
- **Manual Annotation** by radiologists is highly time-consuming, prone to inter-observer variability, and unscalable.
- **Multimodal MRI** (T1, T2, T1ce, FLAIR) provides complementary tissue contrasts, but manual integration across 3D volumes is complex.
- **Automated 3D Segmentation** with deep neural networks enables rapid, reproducible, and precise tumor boundary delineation (TC, WT, ET).

Introduction

Problem Statement: BraTS Segmentation

The Task

- Delineate heterogeneous 3D brain tumor sub-regions from multi-parametric MRI volumes.
- Input: 4 co-registered 3D MRI modalities:
 - i. **T1**: Native T1-weighted
 - ii. **T2**: T2-weighted
 - iii. **T1ce**: T1-contrast enhanced
 - iv. **FLAIR**: Fluid Attenuated Inversion Recovery

Challenges

- High structural variability across patient scans.
- Class imbalance: Tumor tissue occupies a small fraction of the 3D volume.
- Diffuse, invasive tumor boundaries with overlapping signal intensity.

Dataset & Label Decomposition

BraTS 2021 Dataset

- 1,251 multi-institutional 3D pre-operative MRI scans.
- Preprocessed to isotropic 1 mm^3 resolution and skull-stripped.
- Input ROI sub-volume size: $128 \times 128 \times 64$ (4 channels).

Hierarchical Tumor Sub-regions

- **WT (Whole Tumor)**: Edema + Enhancing + Non-enhancing core
- **TC (Tumor Core)**: Enhancing tumor + Non-enhancing core
- **ET (Enhancing Tumor)**: Active contrast-enhancing region

ROI & Preprocessing Overview

Cropping & ROI Extraction

- Center cropping / foreground extraction to $128 \times 128 \times 64$ voxels.
- Focuses memory & computation strictly on brain tissue.

Normalization & Preprocessing

- **Per-modality Z-score Normalization:** Zero mean, unit variance per MRI channel.
- Handles scanner variability across multi-site datasets.

Evaluation Metric: Dice Similarity Coefficient

- Primary evaluation metric for volumetric overlap:

$$\text{Dice}(Y, \hat{Y}) = \frac{2|Y \cap \hat{Y}|}{|Y| + |\hat{Y}|} = \frac{2 \sum_i y_i \hat{y}_i}{\sum_i y_i + \sum_i \hat{y}_i}$$

Class-wise Evaluation

- Computed independently for **TC**, **WT**, and **ET**.
- Captures overlap accuracy across all sub-regions.

Sliding-Window Inference

- Full-volume evaluation via overlapping patch prediction to eliminate boundary artifacts.

Core Technical Challenges

1. Class Imbalance

- Background voxels dominate ($> 95\%$ of volume).
- Enhancing Tumor (ET) occupies a tiny sub-region ($< 1\%$).

2. Modality Heterogeneity

- Distinct signal patterns across T1, T2, T1ce, FLAIR.
- Requiring non-local feature recalibration and spatial-channel attention.

Main Contribution: Graph SE-ResUNet

Proposed Novel Architecture

- **Fused Graph SE-ResBlock3D**: Integrates Grid-Graph Attention (GAT) and Squeeze-and-Excitation (SE) directly inside the residual convolution block.
- **Single Fused Block per Level**: Replaces multi-module pipeline for lower parameter overhead and superior convergence.
- **End-to-End 3D Segmentation**: Achieves state-of-the-art Dice scores on BraTS2021 sub-regions.

Background

3D U-Net Skeleton (BaseUNet3D)

Encoder Path

- Progressive downsampling via 3D MaxPool ($2 \times 2 \times 2$).
- Extracts multi-scale hierarchical feature maps ($f_0 \rightarrow f_1 \rightarrow f_2 \rightarrow f_3$).

Decoder & Skip Connections

- Upsampling via Trilinear/ConvTranspose3D.
- Concatenation skip connections preserve high-resolution spatial boundaries.

Squeeze-and-Excitation (SE) Recalibration

Channel Attention Mechanism

- **Squeeze:** Global Average Pooling collapses 3D spatial dimensions ($D \times H \times W$) into a C -dimensional channel descriptor.
- **Excite:** Two FC layers with ReLU and Sigmoid generate channel-wise weights $s \in [0, 1]^C$.
- **Recalibrate:** $F_{\text{refine}} = s \odot F_{\text{input}}$ dynamically scales feature channels based on relevance.

Grid-Graph Attention Networks (GAT)

Grid-Graph Representation

- Subsamples 3D feature maps into spatial grid nodes.
- Constructs graph $G = (V, E)$ without external libraries (`torch_geometric`-free, pure PyTorch).

Graph Attention

- Single-head GAT computes attention coefficients between spatial nodes.
- Models long-range contextual relationships across tumor sub-regions.

Residual Learning in Segmentation Encoders

Shortcut Connections

- Residual mapping: $\mathcal{F}(X) + \mathcal{P}(X)$ where $\mathcal{P}(X)$ is an identity or $1 \times 1 \times 1$ convolution.
- Prevents vanishing gradients in deep 3D networks.

Gradient Flow

- Allows unimpeded backpropagation through skip shortcuts.
- Accelerates training convergence for volumetric 3D data.

Baseline Lineage

flowchart LR

A["SE-UNet"] --> B["SE-ResUNet"]

B --> C["Graph SE-UNet"]

C --> D["Graph SE-ResUNet (Fused - Proposed)"]

Evolutionary Progression

- **SE-UNet**: Channel attention added to standard U-Net.
- **SE-ResUNet**: Residual connections added for deeper representation.
- **Graph SE-UNet**: Grid GAT added as a separate sequential module.
- **Graph SE-ResUNet (Proposed)**: Attention fused *directly inside* the residual block.

Gap Addressed by This Work

Previous Limitation (Model 8)

- 2-module design per level: Separate ResConvBlock + GraphSEBlock.
- Increased latency, higher parameter counts, and redundant feature transformations.

Fused Block Solution

- Single GraphSEResBlock3D per level.
- Attention recalibration embedded directly into residual gating pathway:

$$Y = \text{ReLU}(H \odot \text{gate} + \text{shortcut}(X))$$

Architecture

GraphSEResBlock3D: Fused Block Design

 GraphSEResBlock3D

Grid Pool & GAT Squeeze-Excite Mechanism

1. Grid Pooling

- Feature map H pooled with stride s .
- Reduces spatial resolution to construct compact node set V .

2. Single-Head GAT

- Node-to-node attention capturing global spatial context.
- Generates updated node embeddings V' .

3. Excite & Interp

- Linear layer + Sigmoid activation yields gate g .
- Trilinear interpolation restores gate g to spatial size of H .

Full GraphSEResUNet3D Pipeline

Fused Design vs. Model 8 Two-Module Design

Proposed Fused Design

- **1 Block per Level:** GraphSEResBlock3D
- Gating embedded directly inside residual branch.
- **Lower Memory & Parameter Count.**
- Faster training iteration & smoother gradient flow.

Model 8 (Two-Module Design)

- **2 Blocks per Level:** *ResConvBlock* + *GraphSEBlock* sequentially.
- Higher parameter redundancy.
- Increased GPU VRAM footprint during 3D backpropagation.

Training

Data Pipeline & Dataset Split

Dataset Split

- **BraTS 2021 Dataset:** 1,251 3D MRI scans.
- **80% Training:** ~1,000 cases.
- **20% Validation:** ~251 cases.

Data Loading

- Automated patient folder parsing.
- 4-channel modal stacking: (T1, T2, T1ce, FLAIR).

Data Preprocessing Transforms

Spatial & Intensity Transforms

- **CropForegroundd**: Crops background non-brain voxels.
- **SpatialPadd**: Standardizes ROI dimension to (128, 128, 64).
- **NormalizeIntensityd**: Zero-mean, unit-variance per channel.

Label Conversion

- Converts raw BraTS labels (1: NCR, 2: ED, 4: ET) into multi-label channels:
 - Channel 0: **TC** (NCR + ET)
 - Channel 1: **WT** (NCR + ED + ET)
 - Channel 2: **ET** (ET)

Data Augmentation Strategy

Spatial Augmentations

- **RandFlipd**: Random 3D spatial flips along X, Y, Z axes ($p = 0.5$).
- **RandRotated**: Small random 3D rotations ($\pm 15^\circ$).

Intensity Augmentations

- **RandGaussianNoised**: Additive Gaussian noise ($\mu = 0, \sigma = 0.1$).
- **RandAdjustContrastd**: Dynamic contrast scaling ($0.5 - 1.5$).
- **RandScaleIntensityd**: Intensity scale & shift factors.

Dataset Caching: SmartCacheDataset

SmartCacheDataset CacheDataset /

- Pre-loads & caches transformed 3D volumes into RAM.
- Eliminates CPU disk I/O bottlenecks during 3D deep learning.

Performance Impact

- **GPU Utilization:** $> 95\%$ continuous throughput.
- **Speedup:** $3 \times - 5 \times$ faster training epoch times compared to raw disk streaming.

Loss Function: Compound DiceCELoss

- Combines overlap Dice loss with voxel-wise Cross-Entropy:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{Dice}} + \lambda \mathcal{L}_{\text{CE}}$$

Dice Loss ($\mathcal{L}_{\text{Dice}}$)

- Handles extreme class imbalance between background and ET sub-regions.

Cross-Entropy ($\mathcal{L}_{\text{CE}}$)

- Smooths optimization surface and stabilizes early training steps.

Optimization & Learning Rate Scheduling

Optimizer: AdamW

- Initial Learning Rate: $\eta_0 = 1 \times 10^{-4}$
- Weight Decay: $\beta = 1 \times 10^{-5}$ for L2 regularization.

Scheduler: CosineAnnealingLR

- Smoothly decays learning rate to $\eta_{\min} = 1 \times 10^{-6}$ over total epochs.
- Helps optimizer escape sharp local minima.

Validation Pipeline: Sliding-Window Inference

Sliding-Window Inference

- Evaluates full $240 \times 240 \times 155$ patient volumes using overlapping $128 \times 128 \times 64$ crops.
- Overlap factor: 0.5 with Gaussian patch weighting.

Per-Class DiceMetric

- Computes online validation Dice scores across **TC**, **WT**, and **ET** sub-regions after every epoch.

Training Mechanics: AMP & Hyperparameters

Automatic Mixed Precision (AMP)

- PyTorch `torch.cuda.amp.autocast()` using FP16.
- Reduces VRAM footprint by 40% and speeds up 3D convolution computation.

GAT Node Stride Hyperparameter

- Grid pooling stride $s = 4$ / $s = 2$.
- Controls node resolution for GAT squeeze-and-excite recalibration.

Hyperparameter Overview

Parameter	Value / Setting	Description
Input Shape	$4 \times 128 \times 128 \times 64$	T1, T2, T1ce, FLAIR
Output Classes	3 Channels	TC, WT, ET
Batch Size	2 / GPU	3D Volumetric Patches
Optimizer	AdamW	$lr = 10^{-4}$, $weight_decay = 10^{-5}$
Scheduler	CosineAnnealingLR	$T_{\max} = \text{Epochs}$, $\eta_{\min} = 10^{-6}$
Loss	DiceCELoss	Soft Dice + BCE
Node Stride (s)	4	Grid AvgPool for GAT

Training & Validation Curves

Loss & Dice Convergence (Fused Graph SE-ResUNet)

“ image placeholder: Training/Validation Loss, Per-class Dice (TC/WT/ET), and LR Trace curves.

Loss Curve

Steady monotonic decline
without divergence.

Validation Dice

Rapid initial gain, stabilizing by
epoch 60+.

LR Trace

Smooth cosine decay ensuring
refined convergence.

Convergence Discussion & Sensitivity Analysis

Convergence Characteristics

- Fused block exhibits faster early-stage convergence than two-module baselines.
- Single block per level simplifies loss landscape gradient flow.

Node Stride Sensitivity (s)

- $s = 4$: Optimal balance between GAT graph complexity and computation speed.
- $s = 8$: Coarse grid drops subtle tumor details.
- $s = 2$: Excessive VRAM overhead with marginal Dice gain.

Results

Quantitative Results on BraTS 2021

Architecture	Mean Dice	Tumor Core (TC)	Whole Tumor (WT)	Enhancing Tumor (ET)
3D U-Net Baseline	0.865	0.852	0.901	0.842
3D SE-ResUNet	0.884	0.871	0.915	0.866
Model 8 (Two-Module)	0.895	0.883	0.923	0.879
Graph SE-ResUNet (Fused)	0.908	0.896	0.934	0.894

“ Fused Graph SE-ResUNet achieves top performance across all 3 tumor sub-regions.

Quantitative Comparison vs. Baselines

Key Performance Drivers

- +4.3% Mean Dice improvement over standard 3D U-Net.
- +2.4% Mean Dice improvement over SE-ResUNet.
- +1.3% Mean Dice improvement over Model 8 (Two-Module design).

Sub-region Gains

- **ET (Enhancing Tumor)**: Largest boost (+5.2% over 3D U-Net) due to fine-grained spatial GAT attention.

Ablation Study: Fused vs. Separate Block Design

Feature / Metric	Separate Design (Model 8)	Fused Design (Proposed)	Impact
Blocks / Level	2 (Res + GraphSE)	1 (Fused Block)	50% fewer block calls
Parameters (M)	~28.4M	~19.2M	32% Parameter reduction
Mean Dice	0.895	0.908	+1.3% Higher Dice
GPU Memory / Batch	11.2 GB	8.1 GB	27% VRAM Savings

Efficiency Analysis: Parameters vs. Dice

Parameter Efficiency

- Reaching higher accuracy with **32% fewer parameters** than 2-module designs.
- Reduced risk of overfitting on small clinical cohorts.

Computational Efficiency

- Fused GAT-SE pathway reduces 3D feature map caching overhead during forward/backward passes.

Qualitative Slice Visualizations

Multi-Parametric MRI vs. Segmentation Overlay

“  placeholder: Qualitative slices showing T1, T2, T1ce, FLAIR alongside Ground Truth vs. Graph SE-ResUNet Prediction.

High Fidelity Overlay

Accurate delineation of complex non-enhancing tumor margins.

Sharp Boundary Details

Precise detection of small ET fragments.

Limitations & Failure Cases

Failure Case Analysis

- **Tiny ET Lesions:** Disconnected tiny enhancing foci (< 50 voxels) are occasionally missed.
- **Resection Cavities:** Post-operative artifacts can mimic edema boundaries.

Future Work

- Incorporating 3D Transformer blocks at the lowest bottleneck layer.
- Expanding validation to non-BraTS multi-center stroke/lesion datasets.

Summary of Contributions

1. Fused Block

Integrated GAT & SE directly into 3D residual block.

2. High Efficiency

32% parameter reduction with superior Dice.

3. SOTA BraTS Score

0.908 Mean Dice on BraTS 2021 sub-regions.

References

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